Isomorphic Questions

Multiple choice questions generated below. Answers in RED and incorrect answers in BLUE.

Q1. Which of the following statements is correct?

A. We should model a bicycle wheel as a point mass when calculating the stress distribution within its spokes.
This is incorrect because internal stresses depend on the wheel's size and shape, so a point-mass model throws away essential details.

B. We can reasonably model a train as a point particle when estimating how far it travels between two stations several kilometers apart over a measured time interval.
This is correct because for kilometer-scale motion the train's dimensions are negligible compared with the displacement, so a point-particle model is appropriate.

C. We are never allowed to treat an object as moving in a vacuum when we want to neglect drag forces.
This is incorrect because "vacuum" is a common idealization used specifically to justify ignoring air resistance when it is small or intentionally neglected.

D. None of the above is correct.
This is incorrect because one of the statements (B) is a valid approximation used in basic motion analysis.

Q2. Which of the following equations is correct?

A. 1 cm = 1 m.
This is incorrect because a centimeter is much smaller than a meter (there are 100 cm in 1 m).

B. 1 = 0.01.
This is incorrect because it equates two pure numbers with different values without any unit conversion context.

C. 1 cm = 0.01 m.
This is correct because 1 centimeter is one-hundredth of a meter.

D. 100 cm = 0.01 m.
This is incorrect because 100 cm equals 1 m, not 0.01 m.

Q3. Which option has the incorrect unit?

A. Power: watt.
This is correct usage because the SI unit of power is the watt (W).

B. Momentum: kg · m/s.
This is correct usage because momentum has units of kg · m/s.

C. Pressure: N/m^3.
This is incorrect because pressure is force per area (N/m^2), not force per volume (N/m^3).

D. Electric charge: coulomb.
This is correct usage because the SI unit of electric charge is the coulomb (C).

Q4. A rescue drone is launched from the base station at the origin and detects a beacon at the coordinates $\vec{r} = 4\hat{\imath} + 7\hat{\jmath}$, where length units are in hectometers. Which of the following statements is correct?

A. The distance between the beacon and the base station is 11 hectometers.
This is incorrect because it adds components $(4+7)$ instead of using the Pythagorean distance.

B. The distance between the beacon and the base station is 7 hectometers.
This is incorrect because it takes only the y-component and ignores the x-component.

C. The distance between the beacon and the base station is 4 hectometers.
This is incorrect because it takes only the x-component and ignores the y-component.

D. None of the above is correct.
This is correct because the distance is $\sqrt{4^2 + 7^2} = \sqrt{65} \approx 8.1$ hectometers, which is not 11, 7, or 4.

Q5. If the displacement of an object is given by the function $\vec{s} = \big((0.80\,\mathrm{m/s^3})t^3 + (1.50\,\mathrm{m/s^2})t^2\big)\hat{\imath}$, which of the following statement is correct?

A. This is a constant acceleration motion.
This is incorrect because the t^3 term makes the acceleration depend on time.

B. The magnitude of the acceleration is 1.50 $\mathrm{m/s^2}$.
This is incorrect because $1.50\,\mathrm{m/s^2}$ is a coefficient in position, not the (constant) acceleration value.

C. The magnitude of the acceleration at $t = 0$ is 0 $\mathrm{m/s^2}$.
This is incorrect because the t^2 term contributes a nonzero constant term to acceleration at $t = 0$.

D. The magnitude of the acceleration at $t = 0$ is 3.0 $\mathrm{m/s^2}$.
This is correct because $a(t) = \frac{d^2s}{dt^2} = (4.8t + 3.0)\hat{\imath}$, so at $t = 0$ the magnitude is 3.0 $\mathrm{m/s^2}$.

Q6. An object takes 4 second to move from position A ($\vec{r}_A = 1.0\hat{\imath} - 2.0\hat{\jmath}$) to position B ($\vec{r}_B = 7.0\hat{\imath} + 1.0\hat{\jmath}$) along a straight path, where length units are in meters. Which of the following statements is correct?

A. The moving distance is 6.7 m.
This is correct because the straight-line distance is $|\Delta\vec{r}| = \sqrt{(7-1)^2 + (1-(-2))^2} = \sqrt{45} \approx 6.7$ m.

B. The speed at $t = 0$ must be 1.7 m/s.
This is incorrect because knowing only start/end positions and total time does not uniquely determine the initial speed without a motion model.

C. The acceleration at $t = 0$ must be 0.42 m/s^2.
This is incorrect because acceleration cannot be inferred from just two positions and a travel time along a straight path.

D. None of the above is correct.
This is incorrect because statement A correctly computes the straight-line distance between the two position vectors.

Q7. A soccer ball is kicked with speed v_0 at an angle θ above the horizontal. What is the speed of the ball when it is at the maximum height (ignore air resistance)?

A. v_0.
This is incorrect because the vertical component decreases to zero at the top, reducing the total speed from its initial value.

B. $v_0 \cos\theta$.
This is correct because at maximum height the vertical velocity is zero while the horizontal component remains $v_0 \cos\theta$.

C. $v_0 \sin\theta$.
This is incorrect because $v_0 \sin\theta$ is the initial vertical component, which is not the speed at the top.

D. 0.
This is incorrect because the projectile still has horizontal motion at the top (unless $\theta = 90°$).

Q8. A robotic submarine starts with an initial velocity of $\vec{v}_0 = (12.0\ \text{m/s})\hat{\imath}$ when its thrusters turn on, producing an acceleration of $\vec{a} = (1.50\ \text{m/s}^2)\hat{\imath} + (0.80\ \text{m/s}^2)\hat{\jmath}$. Based on this scenario, which of the following statements is correct?

A. The x-component of the submarine's velocity remains constant.
This is incorrect because a nonzero a_x means v_x changes with time.

B. The y-component of the submarine's velocity remains constant.
This is incorrect because a nonzero a_y means v_y changes with time.

C. Both the direction of motion and speed change.
This is correct because acceleration has both x and y components, changing both velocity components and thus altering direction and magnitude.

D. None of the above is correct.
This is incorrect because statement C correctly describes the consequence of having both $a_x \neq 0$ and $a_y \neq 0$.

Q9. If the magnitudes of both charges are quadrupled and their separation is also quadrupled, what happens to the magnitude of the electrostatic force that each charge exerts on the other?

A. It stays the same.
This is correct because Coulomb's law gives $F \propto q_1 q_2 / r^2$, so multiplying each charge by 4 multiplies the numerator by 16 while multiplying r by 4 multiplies r^2 by 16, canceling out.

B. It decreases by a factor of $4^{1/2}$.
This is incorrect because it treats the distance dependence as $1/r$ instead of the correct $1/r^2$.

C. It increases by a factor of 4.
This is incorrect because it effectively compares only one charge scaling and ignores that both charges and r^2 change.

D. It increases by a factor of 16.
This is incorrect because it accounts for the increased charges but forgets that increasing the separation reduces the force by the same factor via r^2.

E. It increases by a factor of $4^{1/2}$.
This is incorrect because it reflects a square-root misunderstanding of the inverse-square dependence on distance.

Q10. A spacecraft can escape a planet with radius R and mass M at an escape speed given by $v = \sqrt{\frac{2GM}{R}}$, where G is the universal gravitational constant. Based on this expression for escape speed, which of the following statements is correct?

A. A heavier spacecraft requires a higher speed to escape from the planet.
This is incorrect because the escape-speed formula depends on the planet's M and R, not the spacecraft's mass.

B. A lighter spacecraft requires a lower speed to escape from the planet.
This is incorrect for the same reason: escape speed is independent of the escaping object's mass (ignoring atmosphere/propulsion limits).

C. An object requires a higher speed to escape from a planet with smaller mass and a larger radius.
This is incorrect because smaller M and larger R both reduce M/R, decreasing escape speed.

D. An object requires a higher speed to escape from a planet with greater mass and a smaller radius.
This is correct because $v \propto \sqrt{M/R}$, so increasing M and decreasing R both increase the required escape speed.

Q11. The force $\vec{F} = (2.50\,\text{N})\hat{\imath} + (6.00\,\text{N})\hat{\jmath}$ acts on a particle as it moves along the x axis from $x = -1.2\,\text{m}$ to $x = 3.8\,\text{m}$. What is the work, in joules, done by the force on the particle?

A. 12.5 J.
This is correct because the displacement is $\Delta\vec{r} = (5.0\text{ m})\hat{\imath}$ so $W = \vec{F}\cdot\Delta\vec{r} = (2.50)(5.0) = 12.5$ J.

B. 30.0 J.
This is incorrect because it multiplies the y-component of the force by the x-displacement even though the displacement has no y-component.

C. 5.00 J.
This is incorrect because it confuses the displacement magnitude (5.0 m) with the final work without including the force component along the motion.

D. 42.5 J.
This is incorrect because it effectively adds contributions from both force components as if the particle moved in both x and y directions.

Q12. A 3.2 kg cart initially moving at 1.8 m/s collides with a bumper and rebounds at a speed of 0.9 m/s moving backward. The collision lasts for 0.15 seconds. What is the average force exerted on the cart during the collision?

A. 19.2 N.
This is incorrect because it effectively uses only the rebound speed (or a partial velocity change) rather than the full change from $+1.8$ m/s to -0.9 m/s.

B. 38.4 N.
This is incorrect because it corresponds to using $\Delta v = 1.8$ m/s instead of the correct $\Delta v = 2.7$ m/s in magnitude.

C. 57.6 N.
This is correct because taking the rebound velocity as negative gives $\Delta v = (-0.9 - 1.8) = -2.7$ m/s, so $|F_{\text{avg}}| = m|\Delta v|/\Delta t = 3.2 \cdot 2.7/0.15 = 57.6$ N.

D. None of the above is correct.
This is incorrect because option C correctly applies the impulse-momentum relation to the given numbers.

Q13. A 80 kg catcher is initially at rest. A 90 g baseball traveling at 140 km/hr is caught and sticks in the catcher's glove. What is the speed of the catcher immediately after the catch?

A. 0.044 m/s.
This is correct because momentum is conserved for a catch: $v = \frac{mv_0}{M+m} = \frac{0.090\,(140\ \text{km/hr} \approx 38.9\ \text{m/s})}{80.09} \approx 0.044$ m/s.

B. 0.12 m/s.
This is incorrect because it typically comes from forgetting to convert km/hr to m/s or from dividing by only the catcher's mass incorrectly.

C. 39 m/s.
This is incorrect because it assumes the catcher keeps the ball's speed, ignoring the much larger combined mass after an inelastic catch.

D. 17.5 m/s.
This is incorrect because it reflects mixing up the momentum equation (using masses incorrectly) and failing to account for the catcher+ball moving together.
